

Comparative Demonstration of Coherent-Pluggable Control Architectures for Disaggregated IPoWDM Networks

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Abstract *We demonstrate three IPoWDM coherent-pluggable workflows: hierarchical IP-controller-only management, EDCC-governed dual-controller access with hierarchical SDN coordination, and direct optical-controller access without hierarchical controller, validating responsibilities, access control, and policy enforcement on a live multi-vendor testbed.* ©2026 The Author(s)

Review

IP-over-WDM (IPoWDM) integrates coherent pluggable transceivers directly into IP routers, replacing standalone optical transponders or muxponders and enabling a more compact packet-optical architecture [1-2]. While this reduces footprint and can simplify deployment, it also changes the traditional management boundary between packet and optical transport. The optical transmission function is integrated into the router via coherent pluggables, but its configuration must still comply with optical line system (OLS) constraints such as spectrum allocation, feasibility, and performance requirements. The pluggable thereby forms the optical line interface (OLI) between the router and the OLS.

The required control information is therefore split across IP and optical domains. The IP Software-Defined Networking (SDN) controller typically has direct access to the routers and their pluggables through open management interfaces such as OpenConfig over NETCONF or gNMI [3-5]. However, the optical SDN controller has the line-system knowledge needed to determine feasible optical parameters, including central frequency, output power, operational mode, and path feasibility [6]. This creates a control question that is still not fully resolved: how should pluggable configuration be coordinated when device access and optical-domain expertise reside in different control systems?

The MANTRA architecture addresses this issue through packet-optical SDN coordination and defines two main alternatives: a Dual-SBI model, where both the IP and optical SDN controllers maintain their own southbound interface (SBI) to the router, and a Single-SBI model, where only the IP SDN controller directly interfaces with the router and its coherent pluggables [1]. Previous work has explored hierarchical coordination and

MANTRA-style implementations for these models [7-10], including demonstrations of autonomous service provisioning under a hierarchical SDN control plane [11-12]. Meanwhile, the trade-off between avoiding write conflicts between the IP and optical SDN controllers and enabling meaningful optical-controller involvement remains a key issue for practical deployment.

Coherent-pluggable control is therefore as much an operational responsibility and access-control issue as an interface one. Operators must define which control entity may read or write specific pluggable parameters, how conflicting actions are prevented, and how configuration remains secure and auditable across packet and optical domains. Mechanisms such as the Network Configuration Access Control Model provide fine-grained access control for NETCONF and RESTCONF operations [13]. Related cross-organizational scenarios, such as customer-owned, carrier-controlled pluggables in Data Center Exchange, have been addressed through ownership-aware governance frameworks [14]. The broader question of how the IP and optical SDN controllers should share pluggable access, and how the resulting control architectures compare in terms of complexity, optical-controller involvement, and access control, remains open. These unresolved trade-offs motivate the comparative demo presented in this paper.

Novelty

The demo compares three coherent-pluggable control workflows, summarized in Fig. 1, that span a design space defined by where pluggable-configuration authority resides and how it is coordinated across the IP and optical domains. Each workflow is implemented on the same multi-vendor testbed and exercised under the same service-provisioning scenario, so that their

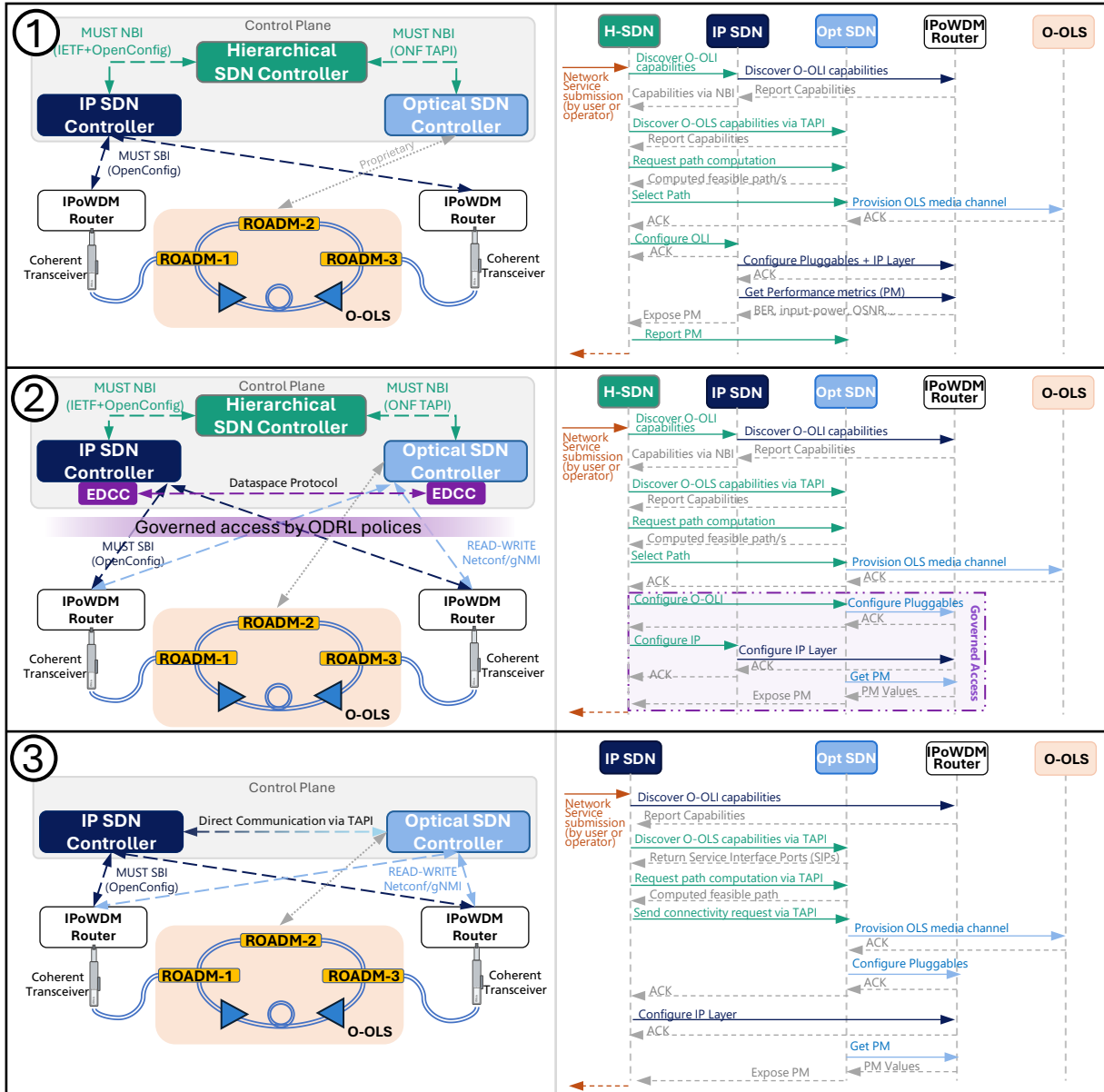


Fig. 1: Three coherent-pluggable control workflows compared in the demo: (a) IP-controller-only router access (MANTRA Proposal 2 [1]); (b) governed dual-controller access via Data Space Connectors, with the hierarchical SDN orchestrator retained; (c) ungoverned dual-controller access with direct east-west coordination between the IP and optical SDN controllers, without a hierarchical SDN controller.

operational differences can be observed directly rather than argued in the abstract.

Workflow 1 in which router access is strictly limited to the IP SDN Controller (MANTRA Proposal 2 [1]) (Fig. 1a). Device access is confined to the IP SDN controller, while the optical SDN controller contributes optical path computation and line-system provisioning through a hierarchical SDN orchestrator. The pluggable is exposed through a single command surface, parallel writes from multiple controllers are avoided by construction, and optical feasibility is validated upstream by the orchestrator before configuration reaches the device. The trade-off is that optical-domain parameters are computed in one domain and applied in another, placing

responsibility for optical correctness on the IP control side and requiring tight coordination through the hierarchical orchestrator.

Workflow 2 in which router is accessed (with read and write rights) by both IP and Optical SDN Controllers in a governed way realized by Data Space Connectors [15-16] (Fig. 1b). The hierarchical orchestrator plays the same role as in workflow 1. Open Digital Rights Language (ODRL)-based YANG augmentation [17] defines which controller may read or write which pluggable parameters, allowing the optical SDN controller to configure permitted optical parameters directly while preventing unauthorized or conflicting modifications by other control entities. This builds on previous work on ownership-aware YANG

models and EDCC-enforced NETCONF operations [14], and on data-space-based governance for secure network data sharing [18-19]. The trade-off is the additional EDCC component on the device path and the operational overhead of authoring and maintaining the policies that govern it.

Workflow 3 in which router is accessed (with read and write rights) by both IP and Optical SDN Controllers. IP and Optical SDN Controllers communicate directly without a hierarchical SDN controller (Fig. 1c). The optical SDN controller is granted read and write access to both the OLS and the pluggable transceivers, allowing it to configure end-to-end optical parameters directly. The IP and optical SDN controllers coordinate east-west, reducing the control stack and avoiding the need to route every interaction through a hierarchical orchestration layer. The trade-off is that conflict prevention and state synchronization must be handled bilaterally between the two controllers, rather than being mediated by an orchestrator or a policy layer.

Demonstrating these three workflows side by side under the same service-provisioning scenario provides a systematic comparison of their practical trade-offs in terms of control-stack complexity, optical-controller involvement, access control, and secure provisioning.

ECOC Relevance

This demo is relevant to ECOC because it addresses a practical control challenge introduced by coherent pluggables in IPoWDM networks. When optical transmission functions are hosted inside IP routers, provisioning is no longer a purely packet-layer or optical-layer task but requires coordinated operation across both domains.

By demonstrating three control workflows side by side on the same testbed, this demo lets the ECOC community observe the trade-offs in control-stack complexity, optical-controller autonomy, access control, and secure multi-domain coordination.

Demo Implementation

The demo validates the three coherent-pluggable control workflows of Fig. 1 under a common IPoWDM service-provisioning scenario. A service request between two IPoWDM routers is triggered, followed by optical path computation, optical line system provisioning, coherent-pluggable configuration, IP/Ethernet-layer configuration, and performance monitoring.

The demonstration will be executed on the HHI testbed in Berlin and remotely controlled from the ECOC Demo Zone. The data plane

consists of two Edgecore SONiC-based IPoWDM routers equipped with Cisco Acacia QSFP-DD 400G ZR+ coherent pluggables, interconnected through a three-node Adtran F8 CloudConnect ROADM-based open optical line system. The control plane includes StableNet® by Infosim as the IP SDN controller and Adtran Mosaic/MNC as the optical SDN controller. EDCC instances are included in workflow 2 to enforce policy-controlled access to pluggable configuration functions.

Each of the three workflows is executed in turn against the same service request. For Workflow 2, the demo uses an example ODRL/YANG policy that permits the optical controller to write the central frequency and output power while restricting other parameters.

```
<odrl-policy xmlns="urn:openconfig:odrl-augmentation">
  <policy-id>pluggable-access-policy</policy-id>

  <permission>
    <id>optical-write-frequency</id>
    ...
    <right-operand>OpticalSDNController</right-operand>
    <right-operand>write</right-operand>
    <right-operand>/components/.../frequency</right-operand>
  </permission>

  <prohibition>
    <id>ip-write-frequency</id>
    ...
    <right-operand>IPSDNController</right-operand>
    <right-operand>write</right-operand>
    <right-operand>/components/.../frequency</right-operand>
  </prohibition>
</odrl-policy>
```

Fig. 2: Shortened ODRL/YANG policy for governed NETCONF access. The policy permits optical-controller writes and prohibits IP-controller writes to the same pluggable frequency target.

Fig. 2 illustrates the policy-enforcement mechanism used in the EDCC-governed workflow. The shown ODRL/YANG policy example grants the optical controller permission to write the frequency parameter, while prohibiting the IP controller from modifying the same parameter. During provisioning, the EDCC verifies controller requests against these policies before the corresponding NETCONF configuration is applied. This realizes responsibility separation at the model level and allows the demo to show both successful and rejected configuration attempts.

Attendees will see each workflow executed through a remote interface that displays the exchanged control messages and configuration steps. After provisioning, monitoring data such as input-power, BER, OSNR, and related performance metrics will be collected to confirm the service state. Attendees can also trigger provisioning interactively and observe how the different workflows affect controller actions, access-control decisions, and conflict prevention.

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References

- [1] O. González de Dios, J. P. Fernández-Palacios Giménez, E. J. Echeverry, S. Melin, R. Diaz, J.-F. Bouquier, J. A. Gómez, M. J. Lopez, D. Breuer, S. Hill, E. Le Rouzic, and A. Mayoral López de Lerma, "TIP MANTRA use cases definition document: IPoDWDM-capable routers' coherent pluggable integration and management," Telecom Infra Project, document version 1.0, Apr. 2024.
- [2] T. Dicorato, P. Landon, P. G. Dicorato, S. Balasundaram, M. Schneeberger, L. Baragiola, and V. Lopez, "Enabling IP-optical integration in core and metro networks," *Journal of Optical Communications and Networking*, vol. 16, no. 7, pp. C121–C132, 2024, DOI: <https://doi.org/10.1364/JOCN.516740>
- [3] R. Enns, M. Björklund, J. Schoenwaelder, and A. Bierman, "Network Configuration Protocol (NETCONF)," IETF RFC 6241, 2011, DOI: <https://doi.org/10.17487/RFC6241>
- [4] OpenConfig, "OpenConfig public YANG models," <https://github.com/openconfig/public>, accessed on 18 May 2026.
- [5] OpenConfig, "gNMI — gRPC Network Management Interface," <https://github.com/openconfig/gnmi>, accessed on 18 May 2026.
- [6] A. Giorgetti, D. Scano, A. Sgambelluri, F. Paolucci, E. Riccardi, R. Morro, P. Castoldi, and F. Cugini, "Enabling hierarchical control of coherent pluggable transceivers in SONiC packet-optical nodes," *Journal of Optical Communications and Networking*, vol. 15, no. 3, pp. 163–173, 2023, DOI: <https://doi.org/10.1364/JOCN.477732>
- [7] A. Mayoral López de Lerma, J.-F. Bouquier, J. A. Gómez, S. Melin, R. Diaz, O. González de Dios, J. P. Fernández-Palacios, K. Coskun, R. Bozaci, S. Hill, V. Augais, D. Breuer, and H. Tuang, "Unified SDN control and management of disaggregated multi-vendor IP over open optical networks," in *European Conference on Optical Communication*, Basel, Switzerland, 2022, paper We2B.1
- [8] O. Gerstel, G. Galimberti, and B. Foster, "A Control Hierarchy for Integrated Packet-Optical Networks Utilizing Pluggable Transceivers," in *Optical Fiber Communication Conference*, San Diego, CA, USA, 2022, paper M4F.2, DOI: <https://doi.org/10.1364/OFC.2022.M4F.2>
- [9] R. Muñoz, L. Gifre Renom, W. Akbar, R. Vilalta, R. Martínez, and R. Casellas, "Dynamic SDN Control of Pluggable Transceivers in disaggregated packet-optical networks for 6G Xhaul," in *International Conference on Transparent Optical Networks*, Barcelona, Spain, 2025, DOI: <https://doi.org/10.1109/ICTON67126.2025.11125125>
- [10] R. Morro, E. Riccardi, D. Scano, A. Sgambelluri, P. Castoldi, F. Cugini, and A. Giorgetti, "First Demonstration of MANTRA IPoWDM Convergent SDN Architecture using SONiC White Box and 400ZR/ZR+ Pluggables," in *International Conference on Optical Network Design and Modeling*, 2023, pp. 1–3.
- [11] H. Zaid, A. Moawad, B. Shariati, R. Emmerich, A. Giorgetti, R. Casellas, C. Schmidt-Langhorst, E. Fernandez, P. Pavon-Marino, F. Cugini, V. Karunakaran, E. Riccardi, R. Morro, R. Bian, C. Matrakidis, C. Santos, P. Gonzalez, L. Velasco, A. Autenrieth, C. Schubert, J. K. Fischer, and R. Freund, "Autonomous Service Provisioning and Self-Healing in Multi-Band Multi-Domain IPoWDM Networks," in *Optical Fiber Communication Conference*, 2025.
- [12] M. R. Raza, A. Moawad, V. Karunakaran, N. Dsilva, A. Sgambelluri, R. Casellas, B. Ali, A. Giorgetti, P. Robles, P. González, V. Tsekenis, S. Barmounakis, P. Demestichas, L. Velasco, F. Cugini, B. Shariati, J. Fischer, and R. Freund, "Demonstration of Self-Healing in IPoWDM Network with Dual Protection Using MBoSDM and NetDevOps Solutions in Support of 6G," in *Optical Fiber Communication Conference*, 2026.
- [13] A. Bierman and M. Björklund, "Network Configuration Access Control Model," IETF RFC 8341, 2018, DOI: <https://doi.org/10.17487/RFC8341>
- [14] A. Mitrovska, H. Zaid, B. Shariati, P. Safari, J. Fischer, and R. Freund, "Demonstration of Customer-Owned and Carrier-Controlled Coherent Pluggables using Ownership-Aware YANG Models," in *Optical Fiber Communication Conference*, Los Angeles, CA, USA, 2026, paper M3Z.1.
- [15] International Data Spaces Association, "IDS Reference Architecture Model (RAM) v4," 2022, <https://internationaldataspaces.org/offers/reference-architecture>, accessed on 28 May 2026.
- [16] Eclipse Foundation, "Eclipse Dataspace Components Connector," <https://projects.eclipse.org/projects/technology.edc>, accessed on 26 May 2026.
- [17] R. Iannella and S. Villata, "ODRL Information Model 2.2," W3C Recommendation, Feb. 2018, <https://www.w3.org/TR/odrl-model/>, accessed on 26 May 2026.
- [18] A. Mitrovska, B. Shariati, A. Jafari, P. Safari, J. K. Fischer, and R. Freund, "Network data sharing: a governance framework for ensuring data sovereignty and privacy compliance," *Journal of Optical Communications and Networking*, vol. 17, no. 11, pp. 1019–1030, 2025, DOI: <https://doi.org/10.1364/JOCN.559523>
- [19] B. Shariati, H. Qarawlus, S. Biehs, J.-J. Pedreno-Manresa, P. Safari, M. Balanici, A. Bouchedoub, H. Haße, A. Autenrieth, J. K. Fischer, and R. Freund, "Telemetry Framework with Data Sovereignty Features," in *Optical Fiber Communication Conference*, 2023, paper M3G.2, DOI: <https://doi.org/10.1364/OFC.2023.M3G.2>